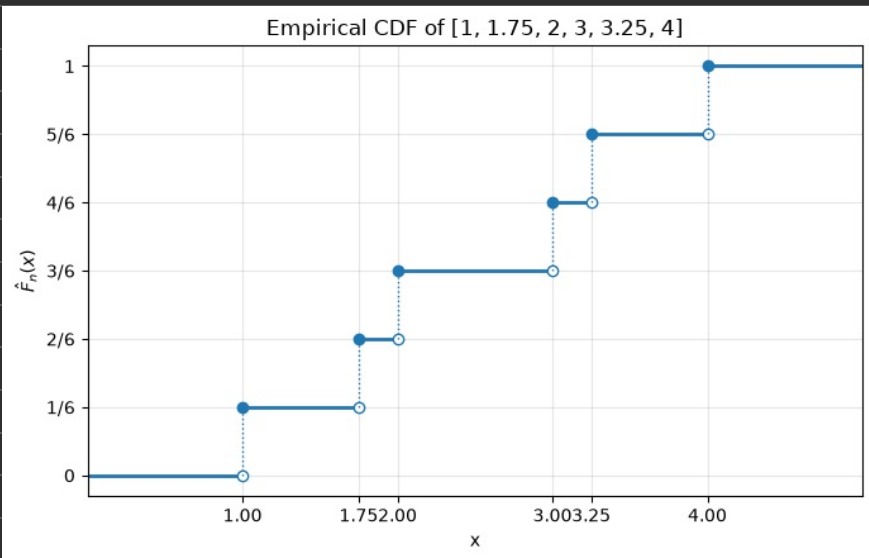


Shuvam Chowdhury
Problem set 5

kernel density estimation

1a) $n=6$

X range	$\hat{F}_n(x)$
$x < 1$	0
$1 \leq x \leq 1.75$	$1/6$
$1.75 \leq x \leq 2$	$2/6$
$2 \leq x \leq 3$	$3/6$
$3 \leq x \leq 3.25$	$4/6$
$3.25 \leq x \leq 4$	$5/6$
$x \geq 4$	1

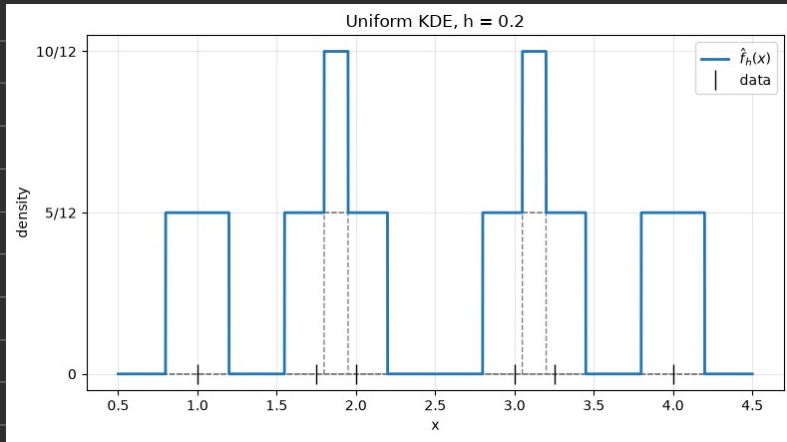


$$b) \hat{f}_h(x) = \frac{1}{nh} \sum_i K\left(\frac{x-x_i}{h}\right)$$

$$\text{width} = 2h = 0.4$$

$$\text{height} = \frac{1}{2} \times \frac{1}{nh} = \frac{1}{2} \times \frac{1}{6 \times 0.2} = \frac{5}{12}$$

x interval	$\hat{f}(x)$
$[0.8, 1.2]$	5/12
$[1.55, 1.8]$	5/12
$[1.8, 1.95]$	10/12 (overlap)
$[1.95, 2.2]$	5/12
$[2.8, 3.05]$	5/12
$[3.05, 3.2]$	10/12 (overlap)
$[3.2, 3.45]$	5/12
$[3.8, 4.2]$	5/12
everywhere else	0



c) $\text{mean} = (1 + 1.75 + 2 + 3 + 3.25 + 4) / 6$
 $= 2.5$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	-1.5	2.25
1.75	-0.75	0.5625
2	-0.5	0.25
3	0.5	0.25
3.25	0.75	0.5625
4	1.5	2.25
Sum	0	6.125

$$\text{variance} \rightarrow SD^2 = \frac{6.125}{6} = 1.0208$$

$$SD = \sqrt{1.0208} \approx 1.0104$$

Plugged in bandwidth formula

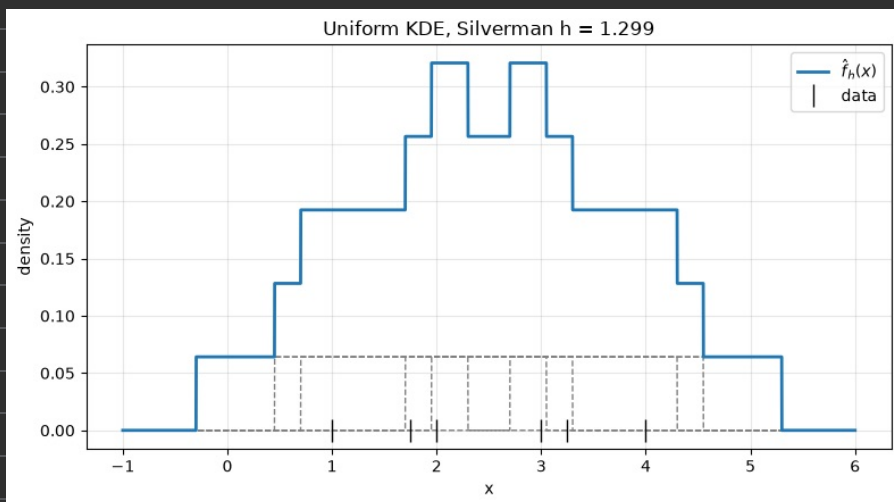
$$h = 1.84 \times 1.0104 \times 6^{(-0.2)} \approx 1.299$$

d) width $= 2h = 2 \times 1.299 \approx 2.598$

$$\text{height} = \frac{1}{2nh} = \frac{1}{2 \cdot 6 \cdot 1.299} = 0.0641$$

x interval	boxes covered	$\hat{f}_n(x)$
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$[-0.299, 0.45]$	1	0.064
$[0.45, 0.701]$	2	0.128
$[0.701, 1.701]$	3	0.192
$[1.701, 1.95]$	4	0.257
$[1.95, 2.299]$	5	0.321
$[2.299, 2.701]$	4	0.257
$[2.701, 3.049]$	5	0.321
$[3.049, 3.299]$	4	0.257
$[3.299, 4.299]$	3	0.192
$[4.299, 4.545]$	2	0.128
$[4.545, 5.299]$	1	0.064
everywhere else	0	0



e) Both use the uniform kernel so both place a box of width $2h$ and height $1/(2nh)$ on every data point and adds them up

but the width of each box is (b) is wider than each box in (d), each box in (b) is taller as well.

The bandwidth controls the bias-variance trade off. The estimates of b is under smoothed. It has low bias but high variance, so it follows each point and gives zero density to gaps that exist only because the sample is small. The estimates in (d) is over smoothed since it has low variance but high bias so it spreads density beyond the range of the data and nearly hides the 2 clusters near 1-2 and 3-4.

$$2a) \hat{f}_h(x) = \frac{1}{2nh} \sum_i \{ |x - x_i| \leq h \}$$

$$n=5, h=0.6 \quad \text{and} \quad x=0$$

$$\hat{f}_{0.6}(0) = \frac{3}{2 \cdot 5 \cdot 0.6} = 0.5$$

$$b) \hat{f}_h(0) = \frac{1}{2nh} \sum_i \{ |x_i| \leq h \}$$

$$E[\hat{f}_h(0)] = \frac{1}{2nh} \sum_i P(|x_i| \leq h)$$

$$= \frac{1}{2nh} \cdot n \cdot P(|x| \leq h)$$

$$P(-0.6 \leq X \leq 0.6) = \Phi(0.6) - \Phi(-0.6) \\ = 2(0.7257) - 1 = 0.4515$$

$$E[\hat{f}_{0.6}(0)] = \frac{0.4515}{2h} \approx 0.376$$

$$c) \varphi(x) = \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{x^2}{2}\right)}$$

$$\varphi(0) = 0.3989$$

d) No they are not equal.

$$\begin{aligned}\text{Bias} &= E[\hat{f}_h(0)] - f(0) = 0.376 - 0.399 \\ &= -0.023\end{aligned}$$

This means KDE is a biased estimator

e) with $h = 0.1$

$$\begin{aligned}P(-0.1 \leq x \leq 0.1) &= 2\Phi(0.1) - 1 = 0.07966 \\ E[\hat{f}_{0.1}^1(0)] &= \frac{0.07966}{2 \cdot 0.1} = 0.3983\end{aligned}$$

as h shrinks the gap between $E[\hat{f}_h(0)]$ and $f(0)$ shrinks towards 0 $\therefore$ the bias disappears as $h \rightarrow 0$.

A narrower window averages the density over a smaller region where the density is nearly flat and close to $f(0)$.

f) Yes $\hat{f}_n$ is consistent, provided h shrinks with n at the right rate.

An estimator is consistent if its mean squared error goes to 0 and

$$MSE = \text{bias}^2 + \text{variance}$$

From (b)-(c) $E[\hat{f}_n(x)]$ is the average of f over $[x-h, x+h]$, and Bias $\approx (h^2/6) f''(x)$. This depends on h , not on n so it goes to 0 only if $h \rightarrow 0$

For variance the number of points in the window is Binomial (n, p) with

$p \approx 2hf(x)$, so $\text{var}[\hat{f}_n(x)] \approx \frac{f(x)}{2nh}$.

This goes to 0 only if $nh \rightarrow \infty$

$\therefore$ the window still captures more and more points even as it narrows

g) the main considerations are

1) The bias variance trade off.

A small h gives low bias but high variance.

A large h gives high bias, low variance.

2) The sample size n since more data supports a smaller h .

3) The spread and scale of the data.
which is why Silverman's rule
scales it by the SD.

3a)
$$\hat{f}_h(x) = \frac{1}{2nh} \sum_i \mathbb{1}\{|x_i - x| < h\}$$

$n = 5$ and $h = 1.5 \therefore 2h = 3$

x	window	Points inside	$\hat{f}_h(x)$
6	$(4.5, 7.5)$	3	$3/15 = 0.2$
6.5	$(5, 8)$	2	$2/15 = 0.133$
7	$(5.5, 8.5)$	3	0.2
8	$(6.5, 9.5)$	3	0.2
8.5	$(7, 10)$	1	0.067

$$f_n^1(n) = \frac{1}{nh} \sum_i \varphi\left(\frac{x-x_i}{h}\right)$$

$$\text{where } \varphi(z) = \frac{1}{\sqrt{2\pi}} e^{\left(\frac{-z^2}{2}\right)}$$

$$n=5, h=1.5 \Rightarrow nh=7.5$$

$$\text{for } x=6.0$$

x_i	$x-x_i$	$z = (x-x_i)/1.5$
5	$6-5=1$	0.667
7	$6-7=-1$	-0.6667
11	11	11
8	$6-8=-2$	-1.3333
10	$6-10=-4$	-2.6667

$$z = \pm 0.6667 = 0.3194$$

$$z = -1.3333 = 0.1640$$

$$z = -2.6667 = 0.0114$$

$$\begin{aligned} \sum \varphi(z_i) &= 0.3194 + 0.3194 + 0.3194 \\ &\quad + 0.1640 + 0.0114 = 1.1338 \end{aligned}$$

x	$x_i = 5$	$x_i = 7$	$x_i = 7$	$x_i = 8$	$x_i = 10$	$f_n(x) = \frac{\text{sum}}{75}$
6	0.3194	0.3194	0.3194	0.164	0.0014	0.151
6.5	0.2420	0.3774	0.3774	0.2420	0.0262	0.169
7	0.1640	0.3989	0.3989	0.3194	0.0540	0.178
8	0.0540	0.3194	0.3194	0.3989	0.1640	0.167
8.5	0.0262	0.2420	0.2420	0.3774	0.2420	0.151

c) The biggest jump in (a) is between $x=6.0$ and $x=8.5$ where Uniform KDE drops from $3/15 = 0.200$ to $1/15 = 0.067$ which is a fall of 0.133 . over the same step the gaussian KDE only goes from 0.167 to 0.151

The uniform kernel has a flat top and vertical edges. Its weight is all or nothing. Inside the window moving x does nothing because every point counts fully no matter where it sits. At $x = 8.5$ both 7s hit the edge and drop out completely and 10 is exactly at edge so it has not yet been counted. The estimate loses 2 points at once and $\therefore$ produces the cliff.

The gaussian kernel does not jump since it is a continuous bell curve with no edges. Its weight decreases gradually with distance

and never reaches zero. for
the same step each 7 moves
from $z=0.67$ to $z=1$ so its
weight decreased from 0.319 to
0.242

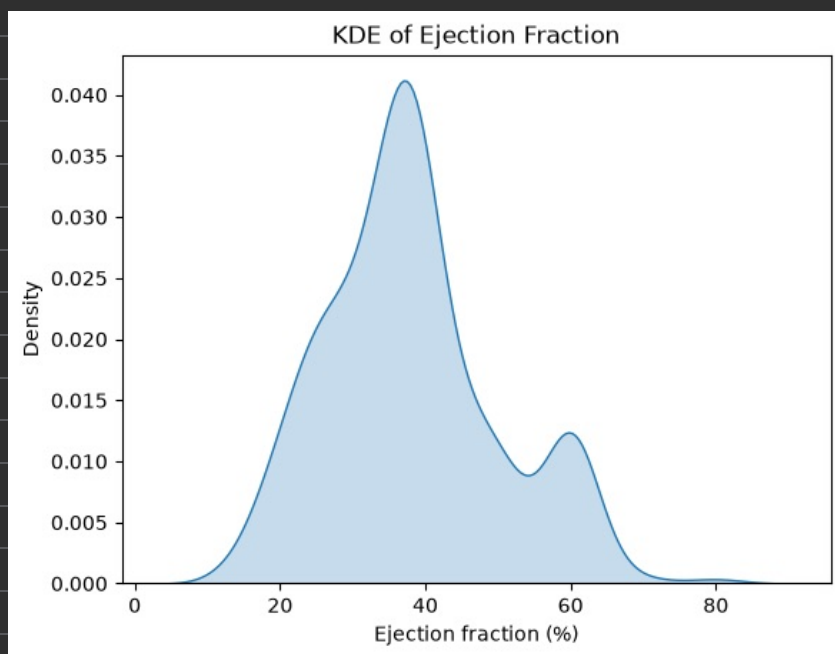
8 moves from $z=0$ to $z=0.33$
so the weight decreases from
0.399 to 0.377

10 moves from $z=-1.33$ to $z=-1$
so weight increases from 0.164 to
0.242

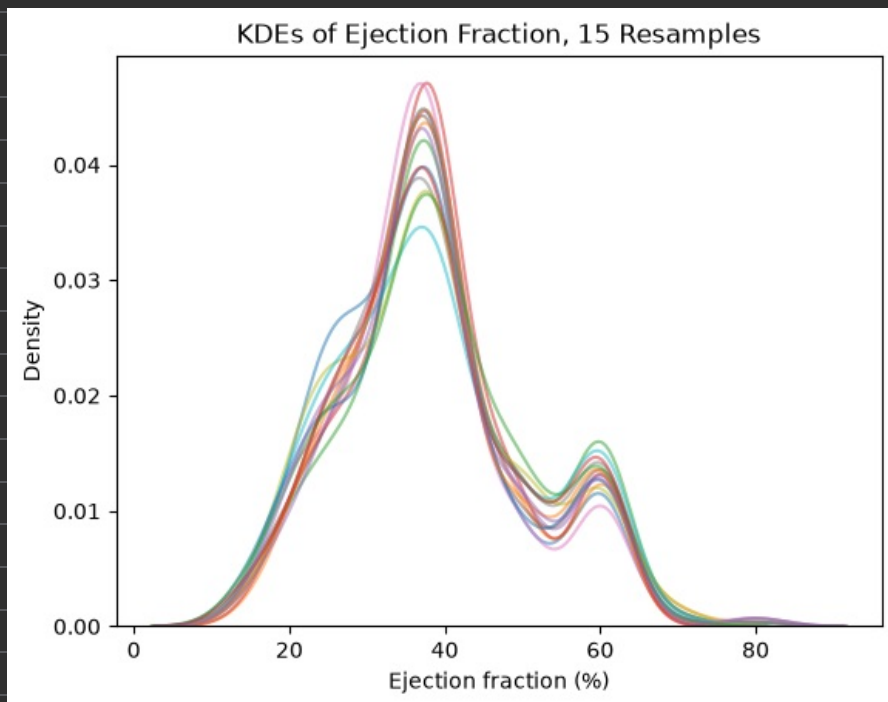
points fade out slowly as x moves
away from them and fade in as x
approaches and the gain from 10
partly offsets the losses from
the 7s.

2) Gaussian kernel is preferred when we want a realistic picture of the density. It is continuous so the KDE is smooth and doesn't jump when point crosses a window edge. Small shifts in the data changes the estimate only slightly. Uniform kernel is preferred when simplicity and speed matters.

4a)



b)



c) The original KDE is reliable below about 15% and above about 70% and on the slopes from about 30% to 35% and 40% and 50% since the 15 resampled curves nearly overlap there. The most noticeable variation is at the main peak at 37% where the resampled heights range from about 0.035 to 0.047. smaller variation shows up in the shoulder near 25% and in the bump near 60%.